

January 7, 2026

(Lecture 3)



Remarks 1: If Ω is discrete and contains at most $n(< \infty)$ points, each single point set $\{\omega_j\}$, $j = 1, 2, \dots, n$, is an elementary event, and it is sufficient to assign probability to each $\{\omega_j\}$. Then if $E \in \mathcal{F}$, where $\mathcal{F}$ is the class of all subsets of Ω , $P(E) = \sum_{\omega \in E} P\{\omega\}$. One such assignment is the *equally likely* assignment or the assignment of uniform probabilities. According to this assignment, $P\{\omega_j\} = 1/n$, $j = 1, 2, \dots, n$. Thus $P(E) = m/n$ if E contains m elementary events, $1 \leq m \leq n$.

Remark 2: If Ω is discrete and contains a countable number of points, one can't make an equally likely assignment of probabilities. It suffices to make the assignment for each elementary event. If $E \in \mathcal{F}$, where $\mathcal{F}$ is the class of all subsets of Ω , define $P(E) = \sum_{\omega \in E} P\{\omega\}$.

Remark 3: If Ω contains uncountably many points, each one-point set is an elementary event, and again one cannot make an equally likely assignment of probabilities. Indeed, one cannot assign positive probability to each elementary event without violating the axiom $P(\Omega) = 1$. In this case one assigns probabilities to compound events consisting of intervals. For example, if $\Omega = [0, 1]$ and $\mathcal{F}$ is the Borel σ -field of all subsets of Ω , the assignment $P[I] = \text{length of } I$, where I is a sub-interval of Ω , defines a probability.

Example 1: Let $\Omega = \{0, 1\}$ and $\mathcal{F}$ is the σ -field of all subsets of Ω . Suppose we define P on $\mathcal{F}$ as follows:

$$P\{1\} = 1/2, \quad P\{0\} = 1/2.$$

Then P clearly defines a probability. Similarly, $P\{1\} = 2/3$, $P\{0\} = 1/3$, and $P\{1\} = 1$, $P\{0\} = 0$, are probabilities defined on $\mathcal{F}$. Indeed, $P\{1\} = p$, $P\{0\} = 1 - p$, ($0 < p < 1$) defines a probability on $(\Omega, \mathcal{F})$.

Example 2: Let $\Omega = \{1, 2, 3, \dots\}$ be the set of positive integers, and let $\mathcal{F}$ be the class of all subsets of Ω . Suppose we define P on $\mathcal{F}$ as follows:

$$P\{i\} = 1/2^i, \quad i = 1, 2, \dots$$

Then $\sum_{i=1}^{\infty} P(i) = \sum_{i=1}^{\infty} 1/2^i = 1$, and P defines a probability.

Example 3: Let $\Omega = \{0, \infty\}$, and let $\mathcal{F} = \mathcal{B}$ be the Borel σ -field on Ω . Suppose we define P on $\mathcal{B}$ as follows:

For each interval $I \subseteq \Omega$,

$$P\{I\} = \int_I e^{-x} dx.$$

Then $P(I) \geq 0$, $P(\Omega) = 1$, and P is countably additive by properties of integrals. Clearly, P proves to be a probability function.

Some Important Properties of the Probability Function:

1. $P(\phi) = 0$.

Proof. Let $E_1 = \Omega$, $E_2 = E_3 = \dots = \phi$. Then using Axiom (ii), we have

$$\begin{aligned} P(\Omega) &= P(E_1) + P(E_2) + P(E_3) + \dots \\ &= P(\Omega) + P(\phi) + P(\phi) + \dots \end{aligned}$$

Since $P(\Omega) = 1$ [from Axiom (iii)], Axiom (i) implies that $P(\phi) \geq 0$, we must have $P(\phi) = 0$.

2. For any finite collection $E_1, E_2, \dots, E_n$ of pairwise disjoint sets in $\mathcal{F}$,

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i).$$

Proof. Using Axiom (ii) and considering the events $E_{n+1}, E_{n+2}, \dots$ as empty sets, we have

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{i=1}^n E_i\right). \quad (1)$$

Also, we have

$$\begin{aligned} \sum_{i=1}^{\infty} P(E_i) &= \lim_{m \rightarrow \infty} \sum_{i=1}^m P(E_i) \\ &= \sum_{i=1}^n P(E_i) + \lim_{m \rightarrow \infty} \sum_{i=n+1}^m P(E_i) \\ &= \sum_{i=1}^n P(E_i) + \lim_{m \rightarrow \infty} 0 \\ &= \sum_{i=1}^n P(E_i). \end{aligned} \quad (2)$$

The result can now be obtained using Axiom (ii) and (1)-(2).

3. P is a monotone function, i.e., if $E_1, E_2 \in \mathcal{F}$ and $E_1 \subset E_2$, then $P(E_1) \leq P(E_2)$.

Proof. We can write E_2 as

$$E_2 = E_1 \cup (E_2 - E_1).$$

This implies that

$$P(E_2) = P(E_1) + P(E_2 - E_1) \text{ [using Axiom (ii)]}. \quad (3)$$

Now since $P(E_2 - E_1) \geq 0$ [using Axiom (i)], $P(E_2) \geq P(E_1)$.

4. For any $E \in \mathcal{F}$, $0 \leq P(E) \leq 1$.

Proof. Since $E \subset \Omega$, $P(E) \leq P(\Omega) = 1$. Also $P(E) \geq 0$ from Axiom (1). This proves the result.

5. $P(E^C) = 1 - P(E)$

Proof. Since $E \cup E^C = \Omega$, we have $P(E \cup E^C) = P(\Omega)$. This implies that $P(E) + P(E^C) = 1$ and the result follows.

6. **Law of Addition:** If $E_1, E_2 \in \mathcal{F}$, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

Proof. We can write

$$E_1 \cup E_2 = E_1 \cup (E_2 - (E_1 \cap E_2)). \quad (4)$$

This implies that

$$P(E_1 \cup E_2) = P(E_1) + P(E_2 - (E_1 \cap E_2)). \quad (5)$$

Also from (3), we have

$$P(E_2 - E_1) = P(E_2) - P(E_1), \text{ if } E_1 \subset E_2.$$

Here since $(E_1 \cap E_2) \subset E_2$, therefore

$$P(E_2 - (E_1 \cap E_2)) = P(E_2) - P(E_1 \cap E_2). \quad (6)$$

The result now follows by substituting (6) in (5).

7. If $E_1, E_2, E_3 \in \mathcal{F}$, then

$$\begin{aligned} P(E_1 \cup E_2 \cup E_3) &= P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) \\ &\quad - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3). \end{aligned}$$

8. **Principle of Inclusion-Exclusion:** If $E_1, E_2, \dots, E_n \in \mathcal{F}$, then

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(E_i) - \sum_{i < j}^n P(E_i \cap E_j) + \sum_{i < j < k}^n P(E_i \cap E_j \cap E_k) \\ &\quad - \dots + (-1)^{n+1} P\left(\bigcap_{i=1}^n E_i\right). \end{aligned}$$